INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

MASK, OXYGEN, MBU-10/P

This specification is approved for use by the Department of the Air Force, and is available for use by all Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of pressure breathing oxygen mask that can be rapidly donned over a headset-microphone. The mask is designated MBU-10/P.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-A-1034	Compressed Air, Breathing
BB-N-411	Nitrogen, Technical
ZZ-R-765	Rubber, Silicone

MILITARY

MIL-B-543	Buckles, Tongueless and Web Strap
MIL-C-5040	Cord, Nylon

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: The Resources and Logistics Services, SA-ALC/TILDM, Kelly AFB TX 78241-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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MIL-W-6858	Welding, Resistance: Spot and Seam
MIL-F-7179	Finishes Coatings and Sealants for the Protection of Aerospace Weapons Systems
MIL-T-8363	Tape and Webbing, Textile, Woven, Nylon
MIL-O-27210	Oxygen, Aviator's
MIL-M-27274	Mask, Oxygen MBU-5/P
MIL-V-27296	Valve, Oxygen Mask, Combination Inhalation and Exhalation
MIL-C-83501	Cord Assemblies, Electrical Braided and Molded Jacket, General Specification for
MIL-C-83867	Connector, Oxygen Mask Hose Non-Ejection Type
MIL-M-87163	Mask, Oxygen MBU-12/P

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-601	Rubber: Sampling and Testing

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-781	Reliability Testing for Engineering Development Qualification, and Production
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards and Specifications, Order of Precedence for the
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements
MIL-STD-2219	Fusion Welding for Aerospace Applications

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

TECHNICAL MANUAL

15x5-3-6-1	Operation, Fitting, Inspection and Instruction with IPB

(Application for copies should be addressed to SA-ALC/TIRT, Kelly AFB, TX 78241-5000)

DRAWINGS

AIR FORCE

60D3519	Face Form - Oxygen Mask, Size Regular Narrow

(Application for copies should be addressed to SA-ALC/TILDB, Kelly AFB, TX 78241-5000)

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2.2 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 4673	Standard Specification for Acrylonitrile - Butadlene - Styrene (ABS) Molding and Extrusion Materials
ASTM D 624	Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017)

SCOTT AVIATION

358-1301	Mask, Oxygen, Quick-Donning Assembly
358-1321	Cord Assembly, Lower
358-1330	Cable and Plug Assembly
358-643	Hanging Suspension Strap

(Application for copies should be addressed to SA-ALC/TILDB, Kelly AFB, TX 78241-5000)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this documents takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4 .

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. The materials shall conform to the applicable specifications and shall be as specified herein. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality and suitable for the purpose intended. Materials that will come in contact with oxygen shall be entirely free from oil, grease, or any other combustible materials.

3.3.1 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt spray, or any atmospheric conditions likely to be met in storage or normal use.

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3.3.1.1 Dissimilar metals. Unless otherwise specified herein, dissimilar metals as defined in MIL-STD-889 having metal-to-metal contact shall not be acceptable. If necessary, combinations of dissimilar metals that have relatively weak galvanic potential may be used provided that their contact surfaces are separated and protected by a passive coating. Any dissimilar metal and protective coating combination proposed to be used shall be presented, prior to fabrication or installation, to the procuring activity for acceptance.

3.3.1.2 Protective treatment. If metals that are subject to deterioration from climatic and environmental conditions likely to occur during service are used, the metals shall be protected in accordance with MIL-F-7179 against this deterioration in a manner that will not in any way prevent compliance with the performance requirements specified herein. The protective coating shall not crack, chip, or scale with age or extremes of climatic and environmental conditions.

3.3.2 Nonmetallic materials. Nonmetallic materials shall be of the ozone-resistant type. Materials that are not susceptible to fungus attack shall be used to the greatest extent possible. Materials that are susceptible to fungus attack, if used, shall be treated with a fungicidal agent.

3.3.2.1 Silicone rubber. The material for the facepiece, the delivery tube, and microphone cable guides shall be silicone rubber. The silicone rubber shall conform to class 3a, grade 50 of ZZ-R-765 except for the brittle point and the hardness. The minimum brittle point of the silicone rubber shall be -56° centigrade (C). The shore A hardness of the silicone rubber shall be 45 $\pm$5. The color of the silicone rubber shall be green, approximately matching 34079 of FED-STD-595. The silicone rubber shall be smooth and free from flash, foreign materials, ventilated forced air oven. The date (year and quarter) the silicone rubber was cured shall be molded on the facepiece and on the delivery tube.

3.3.2.2 Plastic. The material for the hardshell shall conform to ASTM D 4673 and shall be green, approximately matching 34079 of FED-STD-595.

3.3.2.3 Nylon cord. The nylon cord for the cord and snap assembly shall conform to type IA of MIL-C-5040.

3.3.2.4 Webbing. The webbing for the suspension assembly shall conform to type V of MIL-T-8363 and shall be green, approximately matching 34079 of FED-STD-595.

* 3.4 Design and construction. The MBU-10/P, Quick Don Mask, is designed for use with USAF standard H-157 headsets. The unit consists of an oxygen mask conforming to MIL-M-27274 or MIL-M-87163, mounted on a specially designed, comfortable, quick-donning, folding metal frame.

3.4.1 Facepiece. The facepiece shall conform to Drawing 60D3519 and to figure 1. The antifreeze chamber and the body of the facepiece shall be of one piece construction as shown on figures 1 and 2. A 1/16 inch diameter hole that is located as shown on figures 1 and 2 shall vent the antifreeze chamber. A removable, leak tight plug shall be installed in the microphone port of the facepiece. The plug may be fabricated from metal, plastic, or hard rubber.

3.4.2 Delivery tube. The delivery tube shall be made of corrugated silicone rubber and shall conform to figure 3. The delivery tube shall have one lower cable guide that conforms to figure 4 and two upper cable guides that conform to figure 5.

3.4.3 Valve. The combination inhalation-exhalation valve shall conform to MIL-V-27296 and shall be installed in the mask as shown on figure 2.

* 3.4.4 Clamp. The delivery tube shall be attached to the combination valve by means of a hose clamp that conforms to MS22064 or other approved hose clamp.

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* 3.4.5 Connector. A connector that conforms to MIL-C-83867 shall be attached to the delivery tube by means of a hose clamp that conforms to MS22064 or other approved hose clamp.

* 3.4.6 Cord and snap assembly. A nylon cord and metal snap assembly shall be installed inside the delivery tube by attaching the nylon cord to the valve retention pin and the metal snap to the connector retention pin as shown in figure 6. The nylon cord shall be as specified (see 3.3.2.3). The metal snap shall be size No. 2/0, 1-3/16 inches in length, 110 pound test (Berkley and Co., IA), or equal.

3.4.7 Hardshell. The hardshell of the mask shall closely follow the contour of the outside of the facepiece and shall be designed to assure compliance with all of the performance requirements specified herein. The edges of the hardshell nearest the user's face shall be flared. All sharp edges of the hardshell shall be rounded.

* 3.4.8 Dust cover. The mask shall be equipped with a dust cover that will prevent dust, dirt, and foreign objects from entering the mask cavity when the mask is not in use. The dust cover shall be made of durable, nonbrittle, lightweight plastic material contoured to rest around the edge of the hardshell and to be retained on the hardshell at the microphone contour. The configuration of the dust cover shall conform to MIL-M-27274 or MIL-M-87163. The dust cover shall not rest on the facepiece contour between the seal area and the body of the facepiece.

3.4.9 Switch. The mask shall have a microswitch that will automatically switch from the boom microphone (see 3.10) to the mask mounted microphone (see 3.11) during the donning of the mask.

3.4.10 Electrical cord assemblies. The mask shall have two electrical cord assemblies: a lower cord assembly that conforms to Drawing 358-1321 and a cable and plug assembly that conforms to Drawing 358-1330. The two electrical cord assemblies shall conform to section 3 of MIL-C-83501 with the following exceptions:

a. The dimensions, the wiring diagram, the electrical connectors, and the insulation color coding of the power cord assembly and of the cable and plug assembly shall conform to section Drawing 358-1321 and to Drawing 358-1330, respectively.

b. The braided jacket for covering the cord shall be composed of black nylon, 210 denier, 5 ends, 24 carri ers, approximately 21 picks per inch.

c. The performance requirements specified (see 3.6.18.1 and 3.6.18.2) shall apply in lieu of the performance requirements described in the paragraphs entitled "Flexing life" and "Flexing "Y" junction in MIL-C-83501.

d. The performance requirements specified (see 3.6.19) shall apply in lieu of the performance requirements described in the paragraph entitled "Isolation" in MIL-C-83501.

e. The performance requirements specified (see 3.6.20.1 and 3.6.20.2) shall apply in lieu of the performance requirements described in the paragraph entitled "Conductor strain relief test" in MIL-C-83501.

f. In the event of any other conflicts between section 3 of MIL-C-83501 and section 3 of this specification, the requirements of section 3 of this specification shall apply.

3.4.11 Hanging strap. The mask shall have a hanging strap that conforms to Drawing 358-643 or an approved equal.

3.4.12 Buckles. The buckles for the suspension assembly shall be double bar buckles that conform to type II, style 2, class 1, size 3/4 inch of MIL-B-543

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3.5 Construction. The mask shall be constructed so that parts will not become loose in service. It shall be built to withstand the strains, jars, vibrations, and any other conditions incident to shipment, storage, installation, and service.

3.5.1 Welding, soldering, and brazing. Welding, soldering, and brazing shall be carefully performed to insure that all joints have proper penetration, proper bead, and proper strength. Welding shall be accomplished in accordance with MIL-W-6858 and/or MIL-STD-2219.

3.6 Performance.

3.6.1 Delivery tube leakage. The delivery tube shall not leak when it subjected to an internal pressure of 5 pounds per square inch (psi) (see 4.6.2).

3.6.2 Mask leakage. The leakage of the inboard facepiece of the mask shall not exceed 100 milliliters per minute when a suction of 13 millimeters of mercury is applied within the facepiece of the mask (see 4.6.3).

3.6.3 Valve installation, inlet performance (low suction). The total leakage through the inlet portion of the combination inhalation-exhalation valve shall be 15 milliliters per minute or less when a suction of 0.19 millimeters of mercury is maintained at the free end of the delivery tube (see 4.6.4).

3.6.4 Valve installation, exhalation performance. A pressure of 20 millimeters of mercury in the facepiece and a pressure between 15 and 19.9 millimeters of mercury in the inlet tubing shall cause the exhalation portion of the combination inhalation-exhalation valve to open (see 4.6.5).

3.6.5 Microphone switching. The suspension assembly shall open and close smoothly, without seizing or binding. Opening of the suspension assembly at an angle of $59^\circ \pm 6^\circ$ as shown on Drawing 358-1301 shall be the only motion that will cause the microswitch to deactivate the boom microphone circuit and to activate the mask mounted microphone circuit. The mask mounted circuit shall remain activated from 65° to a minimum of 90°. This switching action shall not affect the earphone circuit of the headset-microphone H-157.

3.6.5.1 Wearing position. When the suspension assembly is in the open (wearing) position, the mask microphone circuit shall function without interruption and the boom mounted microphone circuit shall be off.

3.6.5.2 Storage position. When the suspension assembly is in the closed (storage) position, the boom micropnone circuit shall function without interruption and the mask mounted microphone circuit shall be off.

3.6.6 Valve installation, inlet performance (high suction). The total leakage through the inlet portion of the combination inhalation-exhalation valve shall be 150 milliliters per minute or less when a suction of 22.4 millimeters of mercury is maintained at the free end of the delivery tube.

3.6.7 Valve installation, exhalation resistance. The exhalation resistance of the mask, at a flow of 100 liters per minute, shall not be greater than 4.67 millimeters of mercury.

3.6.8 Delivery tube strength. The delivery tube shall not tear or separate from the end connections when it is subjected to a straight pull of 63 pounds ± 1, for 3 seconds ± 2. After this pull has been exerted on the delivery tube, the delivery tube shall meet the requirements specified (see 3.6.1).

3.6.9 Nape hanging strap strength. While the mask is supported by its nape hanging strap, the mask shall withstand a force of 20 pounds, applied at the connector end of the delivery tube, without damage to the mask and without interruption of the boom microphone circuit.

3.6.10 Reliability. The suspension assembly shall have a mean cycle between failures (MCBF) of not less than 1,000 cycles at a 90 percent lower confidence limit.

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3.6.11 Maintainability. No corrective maintenance task time for the mask shall exceed 20 minutes of maintenance downtime.

3.6.12 Environmental. The mask, while it is subjected to temperatures in the range of -40° to 160° Fahrenheit (F), shall meet the requirements specified (see 3.6.5).

3.6.13 Subjective use. The mask shall not have any performance characteristics or any other properties that might be detrimental or objectionable for human use.

3.6.14 Tear resistance. The silicone rubber shall have a minimum tear resistance of 150 pounds per inch when tested (see 4.6.15).

3.6.15 Hardness. The facepiece shall have a Shore A hardness of 45 ±5 when tested (see 4.6.16).

3.6.16 Toxicity. The facepiece shall not contain any chemicals or any other materials that have toxic properties (see 4.6.17).

3.6.17 Hardshell flexibility. The mask hardshell shall withstand a decrease in width of 0.50 inch without visible cracking or fracture when subjected to compressive loading (see 4.6.18).

3.6.18 Flexing life.

3.6.18.1 Legs. The legs of the lower cord assembly that conforms to Drawing 358-1321 and the legs of the cable and plug assembly that conforms to Drawing 358-1330 shall meet the following requirements: The legs, 5 inches or more in length with four or more conductors, of the assemblies shall have a mean flex life of 50,000 flexes. All legs, 5 inches or more in length with three or less conductors, of the assemblies shall have a mean flex life of 25,000 flexes. The required number of flexes shall be performed without evidence of damage to the outside jacket, of damage to the insulation of the individual conductors, and of electrical discontinuity.

3.6.18.2 "Y" junction. The lower cord assembly that conforms to Drawing 358-1321 shall have a mean flex life at the "Y" junction of 25,000 flexes. The required number of flexes shall be performed without any evidence of damage or of discontinuity. The cable and plug assembly that conforms to Drawing 353-1330 will not be subjected to this requirement.

3.6.19 Isolation. The lower cord assembly that conforms to Drawing 358-1321, less the connectors, shall meet the isolation requirements listed in table I. The cable and plug assembly that conforms to Drawing 358-1330 will not be subjected to this requirement.

TABLE I. Isolation

Frequency (Hertz)	Electrostatic Isolation (Minimum Decibels)	Electromagnetic Isolation (Minimum Decibels)
20,000	100	69
10,000	106	75
5,000	112	81

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3.6.20 Conductor strain relief with connectors.

3.6.20.1 Cable and plug assembly. The cable and plug assembly that conforms to Drawing 358-1330 shall not show any signs of slippage of the braided jacket or of electrical failure and shall meet the performance requirements specified in the paragraph entitled "Continuity" of MIL-C-83501 when tensile load is applied between the connectors and the cable.

3.6.20.2 Lower cord assembly. The lower cord assembly that conforms to Drawing 358-1321 shall not show any signs of slippage of the braided jacket or electrical failure and shall meet the performance requirements specified in the paragraph entitled "Continuity" of MIL-C-83501 and the performance requirements specified (see 3.6.19) of this specification when tensile load is applied between the connectors and the cable.

3.7 Interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Weight. The weight of the completely assembled mask shall not exceed 2.25 pounds.

3.9 Marking

3.9.1 Identification of product. The facepiece and the suspension assembly shall be marked for identification in accordance with MIL-STD-130.

3.9.1.1 Facepiece. The following specific information shall be marked on the facepiece:

Regular Narrow
Manufacturer
Date of Manufacture (Quarter and Year)

3.9.1.2 Suspension assembly. The following specific information shall be marked on the suspension assembly:

Mask, Oxygen, MBU-10/P
Manufacturer

3.9.2 Delivery tube. The following words shall be molded into the rubber of the mask end of the delivery tube: **MASK END**. The date of manufacture (quarter and year) shall be molded on either end of the delivery tube.

3.10 Government-furnished property. When specified (see 6.2), the following item (mask mounted microphone) will be furnished by the Government:

Item Description	Quantity Required
Microphone, Dynamic M-100/AIC (MBU-5/P Mask)	1
Microphone, Dynamic M-101/AIC (MBU-12/P Mask)	1

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3.11 Government-loaned property. When specified (see 6.2), the following items will be loaned by the Government to the manufacturer of the mask:

Item Description	Quantity
Headset Microphone H-157 consisting of the Headset, Electrical H-158/AIC and the Microphone, Dynamic M-87/AIC	1

3.12 Workman. The mask shall be constructed in accordance with good commercial practice and shall be free from any defects that might affect appearance or function. The mask shall be clean, dry, and free from oil, grease, and any other foreign matter. All metal and plastic parts shall be smooth and free of sharp edges. All excess flash shall be properly trimmed. The facepiece and its related parts shall be smooth and free from blisters, porous areas, cracks, abrasions, bubbles, and foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection condition

4.3.1 Atmospheric conditions. Unless otherwise specified all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of $77^{\circ} \pm 18^{\circ}$ Fahrenheit (F), and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

4.3.2 Gas. The gas used in testing shall be oxygen that conforms to type I of MIL-O-27210. The gas used in all other testing shall be oxygen that conforms to type I of MIL-O-27210; water pumped nitrogen that conforms to type I, grade B, class 1 of BB-N-411; or compressed air that conforms to grade A or C of BB-A-1034. If the gas used is nitrogen or compressed air, appropriate density correction factors shall be applied to the flowmeter used.

* 4.4 First article inspection. First article inspection shall be performed on four complete mask assemblies, when first article sample is required (see 3.1 and 6.2). This inspection shall include the examinations and tests as described herein (see 4.6) before regular production is started.

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4.4.1 Inspection facility. The first article shall be inspected in a laboratory acceptable to the Government or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity. The required number of first articles shall be stated in the contract. The inspections shall be as specified (see 4.6).

4.4.1.1 Acceptance for reliability. The acceptance for the reliability testing methods specified (see 4.6.11) shall be judged on the basis of the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting procedures for the reliability testing shall be in accordance with MIL-STD-781.

4.2 Facepiece. The extra facepiece and the four sheets of silicone rubber shall be subjected to the tests specified (see 4.6.15 thru 4.6.17).

4.4.3 Hardshells. The three extra hardshells shall be subjected to the hardshell flexibility test specified (see 4.6.18).

4.4.4 Cord assemblies. The five extra terminated cord assemblies and the five extra unterminated cord assemblies shall be subjected to the examinations and tests specified for first article inspection in section 4 of MIL-C-83501 with the following exceptions:

a. The test specified (see 4.6.19.1 and 4.6.19.2) in this specification shall apply in lieu of the tests specified under "Flexing life" in MIL-C-83501.

b. The test specified (see 4.6.20) in this specification shall apply in lieu of the test under "Isolation" in MIL-C-83501.

c. The test specified (see 4.6.21.1) in this specification shall apply in lieu of the test specified under "Conductor strain relief cords assembled with connectors" in MIL-C-83501.

4.4.5 First article samples. The first article samples shall be representative of the production equipment and shall be identified with the manufacturer's part number and any other information required by the procuring activity (see 6.3). The samples shall consist of the following items:

a. Four completely assembled masks.

b. One extra facepiece.

c. Three extra hardshells.

d. Four sheets, 6 inches by 6 inches by 0.06 inches, of silicone rubber that is identical in composition and cure to the silicone rubber used in the facepieces.

e. Five extra terminated cord assemblies.

f. Five extra unterminated cord assemblies.

4.4.6 Samples and reports. The following items shall be furnished to the procuring activity (see 6.3):

a. The tested samples.

b. Three copies of the test report that conforms to MIL-STD-831 and that includes the results of all tests and a detailed statement of compliance or noncompliance with each requirement of this specification, identified by the applicable paragraph number.

c. Three copies of a silicone rubber report that shall contain the following:

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(1) Compound number and name of manufacturer.

(2) Composition of compound, including fillers and catalysts, giving parts by weight of each constituent.

(3) Mixing and curing instructions.

(4) Any other information that the manufacturer desires to furnish for better identification of the material or for showing that the material has desirable characteristics for oxygen usage.

4.5 Quality conformance inspection. Unless otherwise specified herein (see 4.5.1), The quality conformance inspection shall consist of the following:

a. Individual tests.

b. Sampling tests.

4.5.1 Electrical cord assemblies. The lower cord assembly and the cable and plug assembly shall be subjected to the quality conformance inspection specified in section 4 of MIL-C-83501 with the following exceptions:

a. The test specified (see 4.6.19.1 and 4.6.19.2) in this specification shall apply in lieu of the test specified under "Flexing life" in MIL-C-83501.

b. The test specified (see 4.6.20) in this specification shall apply in lieu of the test specified under "Isolation" in MIL-C-83501.

c. The test specified (see 4.6.21.1 and 4.6.21.2) in this specification shall apply in lieu of the test specified under "Conductor strain relief cords assembled with connectors" in MIL-C-83501.

d. In the event of any other conflicts between section 4 of MIL-C-83501 and section 4 of this specification, the requirements of section 4 of this specification shall apply.

4.5.2 Individual tests. Each mask shall be subjected to the following method of inspection:

a. Examination of product (see 4.6.1).

b. Delivery tube leakage (see 4.6.2).

c. Mask leakage (see 4.6.3).

d. Valve installation, inlet performance (low suction) (see 4.6.4).

e. Valve installation, exhalation performance (see 4.6.5).

f. Microphone switching (see 4.6.6).

4.5.3 Sampling tests

4.5.3.1 Plan A. Two masks shall be selected at random from each lot of 1,000 masks, or fraction thereof, which have been manufactured essentially under the same conditions and at the same time. Each mask that is selected as a sample shall be subjected to the following method of inspection:

a. Valve installation, inlet performance (high suction) (see 4.6.7).

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b. Valve installation, exhalation resistance (see 4.6.8).

c. Delivery tube strength (see 4.6.9).

4.5.3.2 Plan B. One facepiece shall be selected at random from all of the facepieces that have been manufactured from the same batch (see 6.4) of silicone rubber compound. Each facepiece selected as a sample shall be subjected to the following method of inspection:

a. Tear resistance (see 4.6.15).

b. Hardness (see 4.6.16).

4.5.3.3 Plan C. Four sheets of silicone rubber shall be fabricated from each batch (see 6.4) of silicone compound rubber. Each sheet of silicone rubber shall be 6 inches by 6 inches by 0.06 inches and shall be identical in composition and cure to the silicone rubber used in the facepiece. Each sheet of silicone rubber shall be subjected to the following method of inspection:

a. Tear resistance (see 4.6.15).

b. Hardness (see 4.6.16).

4.5.3.4 Plan D. If the hardshells are not molded in a essentially continuous operation, three hardshells shall be selected at random from all of the hardshells that have been molded under the same conditions, at substantially the same time, and from the same batch (see 6.4) of plastic molding material. Each hardshell that is selected as a sample shall be subjected to the hardshell flexibility test specified (see 4.6.18).

4.5.3.5 Plan E. If the hardshells are molded in a essentially continuous operation three hardshells shall be selected at random from each day's production of each mold cavity. Each hardshell that is selected as a sample shall be subjected to the hardshell flexibility test specified (see 4.6.18).

4.5.3.6 Rejection and retest

4.5.3.6.1 Sampling plans A, B, C, and D. If a sample selected in accordance with plan A, B, C, and D fails to meet the specification, acceptance of all the items represented by the sample shall be withheld until the extent and cause of failure have been determined and corrected. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the cause of failure, the action taken to preclude recurrence, and the impact the failure may have on scheduled deliveries. After correction, all of the sampling tests shall be repeated.

4.5.3.6.2 Sampling plan E. If a sample selected from a production run in accordance with sampling plan E fails to meet the specification, no items still on or later produced shall be accepted until the extent and cause of failure have been determined. The contractor shall explain full to the Government representative and notify the procuring activity in writing the cause of failure, the action taken to preclude recurrence, and the impact this failure may have on scheduled deliveries.

4.5.3.7 Continuation of tests. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the items represented by the samples shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the methods of correcting them.

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4.6 Methods of inspection

4.6.1 Examination of product. The completely assembled mask shall be examined to determine conformance to the requirements of this specification with respect to materials, design, construction, dimensions, markings, and workmanship.

4.6.2 Delivery tube leakage. One end of the delivery tube shall be plugged. An air pressure of 5 psi shall be applied to the other end of the delivery tube. The delivery tube shall be submerged just under the surface of water and observed. Bubbles rising from the delivery tube after it has been submerged 10 to 15 seconds shall be cause for rejection.

4.6.3 Mask leakage. The mask with the combination inhalation-exhalation valve correctly installed shall be placed on a suitable test stand and adjusted so that there is no leakage around the face seal. The free end of the mask tubing shall be plugged. A suction of 13 millimeters of mercury shall be applied within the facepiece of the mask. While this suction is applied, the inboard facepiece leakage shall not exceed 100 milliliters.

4.6.4 Valve installation, inlet performance (low suction). The inlet portion of the combination inhalation-exhalation valve shall be completely dry during the test. A suction of 0.19 millimeters of mercury shall be applied and maintained at the free end of the delivery tube. While this suction is maintained, the inlet portion of the combination inhalation-exhalation valve shall be inspected for imperfect seating of the flapper on the seat and for leaks around the vale seat. The total leakage through the inlet portion of the combination inhalation-exhalation valve shall be not more than 15 milliliters per minute. If necessary, a suitable means to prevent collapse of the hose may be used.

4.6.5 Valve installation, exhalation performance. The mask shall be placed on a suitable test stand. All suction and pressure in the mask from the previous test shall be exhausted. A pressure of 20 millimeters of mercury shall be applied at the mask tubing. The pressure at the mask tubing shall then be reduced gradually, causing the inlet portion of the combination inhalation-exhalation valve to close and causing a pressure differential to be applied across the exhalation portion of the combination inhalation-exhalation valve. The exhalation portion of the combination inhalation-exhalation valve shall open with a pressure in the tube between 15 and 19.9 millimeters of mercury. If extreme pressure is encountered, the inlet portion of the combination inhalation-exhalation valve shall be rechecked for leakage. If the combination inhalation-exhalation valve is properly seated and the mask still fails to function properly, the combination inhalation-exhalation valve shall be replaced, and the test shall be repeated.

4.6.6 Microphone switching. The suspension assembly of the mask shall be operated. The angle at which the microswitch activates shall be determined and shall be as specified (see 3.6.5).

4.6.7 Valve installation, inlet performance (high suction). The inlet portion of the combination inhalation-exhalation valve shall be completely dry during this test. A suction of 22.4 millimeters of mercury shall be applied and maintained at the free end of the delivery tube. While this suction is maintained, the inlet portion of the combination inhalation-exhalation valve shall be inspected for imperfect seating of the flapper on the seat and for leaks around the valve seat.

4.6.8 Valve installation, exhalation resistance. The mask, with the combination inhalation-exhalation valve installed, shall be placed on a suitable test stand. At an exhalation flow of 100 liters per minute, the pressure inside the mask shall not exceed ambient pressure by more than 4.67 millimeters of mercury.

4.6.9 Delivery tube strength. The cord and snap assembly shall be attached to the mask before this test is commenced. One end of the delivery tube shall be attached to the combination inhalation-exhalation valve mounted on the mask body. The other end of the delivery tube shall be connected to a fitting that simulates the connector specified (see 3.4.5). The clamps for the connections shall be of the type specified herein. One of the

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end connections shall be attached to a support. A weight of 63 pounds ± 1 pound shall be attached to the other connection so that a straight pull will be exerted on the tubing for a period of 3 seconds ± 2 seconds. The tubing shall ;not tear and shall not separate from the end connections. Upon completion of this test, the delivery tube shall be subjected to the delivery tube leakage test specified (see 4.6.2).

4.6.10 Nape hanging strap strength. The mask shall be freely supported by the nape hanging strap. A weight of 20 pounds ±0.50 pounds shall be hung from the connector end of the delivery tube. While this force is applied, the boom microphone electrical circuit shall be continuous.

4.6.11 Reliability. The suspension assembly of the mask shall be subjected to repeated cycling. After each 250 cycles, the mask shall be tested as specified (see 4.6.6). Failure shall be identified as the inability of the mask to meet the requirements specified (3.6.5). A cycle shall consist of the following steps:

a. Open the suspension assembly to the wearing position.

b. Close the suspension assembly to the storage position.

4.6.12 Maintainability verification. Time studies of all corrective maintenance tasks for all expected failure modes shall be conducted by simulating failures. Timing of each maintenance task to determine average values shall be performed at least three times. Only maintenance tasks that are related to field level maintenance shall be considered. The time to accomplish each maintenance task shall include time for preparation, recognition, fault correction, adjustment and realignment, final test, and replacement but shall exclude supply time and administrative time. Redetermination of the average time required, in the case of repetitive subtasks that occur by the nature of the sequence of the maintenance actions (e.g. removal), shall not be necessary.

4.6.13 Environmental. The mask shall be subjected to the high temperature and low temperature tests. The mask shall be allowed to reach stabilized conditions after each step, before proceeding to the next step, of the high temperature and the low temperature tests.

4.6.13.1 High temperature. The mask shall be subjected to a high temperature test in accordance with method 501, procedure II of MIL-STD-810. The operating parameters specified (see 3.6.5) shall be used in lieu of the pretest performance record specified in MIL-STD- 810. The temperature for step 7 shall be 160° F. The suspension assembly shall be cycled (see 4.6.11) once every 10 minutes for 4 hours during step 8.

4.6.13.2 Low temperature. The mask shall be subjected to a low temperature test in accordance with method 502, procedure II of MIL-STD-810. The operating parameters specified (see 3.6.5) shall be used in lieu of the pretest performance record specified in MIL-STD-810. The temperature for steps 2 and 4 shall be -40° F. The time period for step 2 shall be 4 hours. The suspension assembly shall be cycled as specified (see 4.6.11) once every 10 minutes for 4 hours during step 4.

4.6.14 Subjective use. The mask shall be used by human subjects for up to 2 hours at altitudes from ground level to 40,000 feet at temperatures from -40° F to 160° F or any combination of such altitude and temperature. Subjective use tests are optional tests to be conducted at the discretion of and by the Government when the mask appears to have performance characteristics or any other properties that might be detrimental or objectionable for human use.

4.6.15 Tear resistance. The tear resistance of the sheets of silicone rubber and of sections of silicone rubber cut from the sealing area of the facepiece shall be determined in accordance with die B of ASTM D 624. The minimum test resistance of the silicone rubber shall be 150 pounds per inch.

4.6.16 Hardness. The hardness of the sheets of silicone rubber and sections of silicone rubber cut from the sealing area of the facepiece in both nose and cheek areas shall be determined in accordance with method 3021 of FED-STD-601. The Shore A hardness of the silicone rubber shall be 45 ±5.

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4.6.17 Toxicity. Toxicity shall be determined from the chemicals used in the formation of the silicone rubber compound.

4.6.18 Hardshell flexibility. The hardshell shall be placed between the heads of a suitable compression testing machine with the heads contacting the sides of the hardshell at the point of maximum width. A suitable means of measuring the decrease in width of the hardshell under load shall be provided. The rate of head travel shall be 0.20 to 0.25 inch per minute. The hardshell shall be deformed until failure or until the required decrease in width is attained, whichever occurs first. The hardshell shall withstand at least 0.5 inch decrease in width without visible cracking or breaking.

4.6.19 Flexing life

4.6.19.1 Legs. All legs 5 inches or more in length, of the lower cord assembly and of the cable and plug assembly shall conform to 3.6.18.1 when tested as follows:

a. Samples of the cable shall be clamped and suspended through holes in a metal bar having a rectangular cross section of 0.50 inches by 0.62 inch. The clamping point shall be immediately above the metal bar; the cord clamp shall rest on top of the metal bar. The cord shall be clamped in such a manner that it will not turn in the hole during the flexing. The through holes in the metal bar shall be .62 inch in length and 0.005 to 0.010 inches in diameter greater than the diameter of the cord which is undergoing test. The through holes in the metal bar shall incorporate a 0.125 inch radius at the end from which the cord emerges (bending point).

b. A 3 pound weight shall be hung from the legs of the assemblies with four or more conductors. A 1 pound weight shall be hung from the legs of the assembles with three or less conductors. The weight shall be at tached at a point 4 to 12 inches below the bar to assure that the cord leg is held in a vertical position

c. Each conductor of each sample cord; a small low current drain 6 volt lamp; and a suitable power source shall be be rotated about its longitudinal axis back and forth through a angle of 120° (occurs shall be taken as the life of the cord.

4.6.19.2 "Y" junction. Each of the two "y" junctions of the lower cord assembly shall conform to 3.6.18.2 when tested in accordance with figure 1 in MIL-C-83501 and as follows:

a. The samples shall be clamped and suspended through holes in a metal bar having a rectangular cross sec tion of 0.50 inch by 0.62 inch. The samples shall be clamped in the metal bar so that the samples will be suspended entirely in the vertical position. The samples shall be positioned so that the "y" joint division is at the point where the cord emerges below the metal bar. The cord shall be clamped in such a manner that it will not turn in the hole during flexing. The through holes in the metal bar shall be 0.62 inch in length and 0.005 to 0.010 inch in diameter greater than the diameter of the cord samples. The through holes in the metal bar shall incorporate a 0.125 inch radius at the side from which the cord emerges (bending point).

b. A length of nylon cord shall be tied to each of the two "y" ends. A 1 pound weight shall be attached open 60° (30° either side of vertical).

c. The metal bar shall be supported horizontally so that it will rotate about the longitudinal axis going through the point where the cord samples emerge from the bar. The metal bar shall be rotated back and forth through an angle of 120° (60° each side of vertical) at a flexing rate of approximately 85 cycles per minute.

d. Each conductor of each sample cord shall have a low current drain monitor circuit (6 VDC source) con nected in series to indicate electrical continuity. The arithmetical mean of the flexing of the samples before electrical discontinuity shall be taken as the flexing life.

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4.6.20 Isolation. The electrostatic isolation of the lower cord assembly shall be determined in accordance with figure 7. The electromagnetic isolation of the lower cord assembly shall be determined in accordance with figure 8. The electrostatic isolation and the electromagnetic isolation of the lower cord assembly shall be within the limits specified in table I.

4.6.21 Conductor strain relief with connectors. The cable and plug assembly shall be subjected to the test specified (see 4.6.21.1). Each end of lower cord assembly shall be subjected to the test specified (see 4.6.21.2)

4.6.21.1 Cable and plug assembly. With the connectors held stationary, the cord shall be gripped at a point at least 3 inches from the connector under test in such a manner that there shall not be any damage to the cord or slippage in the grip. A straight pull in a manner to avoid a sudden jerk on the connector shall be applied to the cable as follows:

a. A pull of 25 pounds shall be applied to the cable in the direction of the axis of the telephone plug U-174/U for a minimum of 2 minutes.

b. A pull of 25 pounds shall be applied to the cable in the direction of the axis of the telephone jack U-173/U for a minimum of 2 minutes.

c. A pull of 8 pounds shall be applied to the cable in the direction of the axis of the plug connector U-173/U for a minimum of 30 minutes.

d. A pull of 8 pounds shall be applied to the cable in the direction of the axis on the connector leading to the automatic switch for a minimum of 30 minutes.

4.6.21.2 Lower cord assembly. With the connector held stationary, the cord shall be gripped at a point at least 3 inches from the connector under test in such a manner that there shall not be any damage to the cord or slippage in the grip. A straight pull of 8 pounds shall be applied to the cable, in a manner to avoid a sudden jerk on the connector, in the direction of the cord connector axis for a minimum of 30 minutes.

4.6.22 Inspection of packaging.

* 5 PACKAGING

5.1 Cleaning and drying. Cleaning and drying of the masks shall be in accordance with 15X5-3-6-1.

* 5.1 Preservation. The desired levels of preservation will be as specified by the acquisition activity in their acquisition document (see 6.2). If requirements are not specified, preservation shall be in accordance with MIL-STD-2073-1.

* 5.2 Packing. The desired levels of packing will be as specified by the acquisition activity in their acquisition document (see 6.2). If requirements are not specified, packing shall be in accordance with MIL-STD-2073-1.

* 5.3 Marking. Marking requirements will be specified by the acquisition activity in their acquisition document. (see 6.2). If requirements are not specified, mark in accordance with MIL-STD-129.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mask covered by this specification is intended to dispense gaseous oxygen to aircrew members during unpressurized emergency descents from altitudes of 40,000 feet or lower.

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* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.).

c. When first article is required (see 3.1, 4.4 and 6.3).

d. Specify type of oxygen mask to be used (see 3.4).

e. Property loaned or furnished by the Government (see 3.10 and 3.11).

f. Selection of applicable levels of preservation, packaging and marking (see 5.1, 5.2 and 5.3).

* 6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to tested as specified in 4.4.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Batch. A batch of silicone rubber compound shall consist of the silicone rubber compound that is manufactured at one time by the manufacturer of the silicone rubber compound. A batch of the plastic molding material shall consist of the plastic molding material that i manufactured at one time by the manufacturer of the plastic molding material.

* 6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.10.

* 6.5.1 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.11.

* 6.6 Subject term (key word) listing.

Batch
Clamp
Connector
Cord, electrical
Cord, nylon
Cover, dust
Facepiece
Hardshell
Plastic
Rubber, silicone
Switch
Tube, delivery
Valve
Webbing

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6.7 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations and relationship to the last previous issue.

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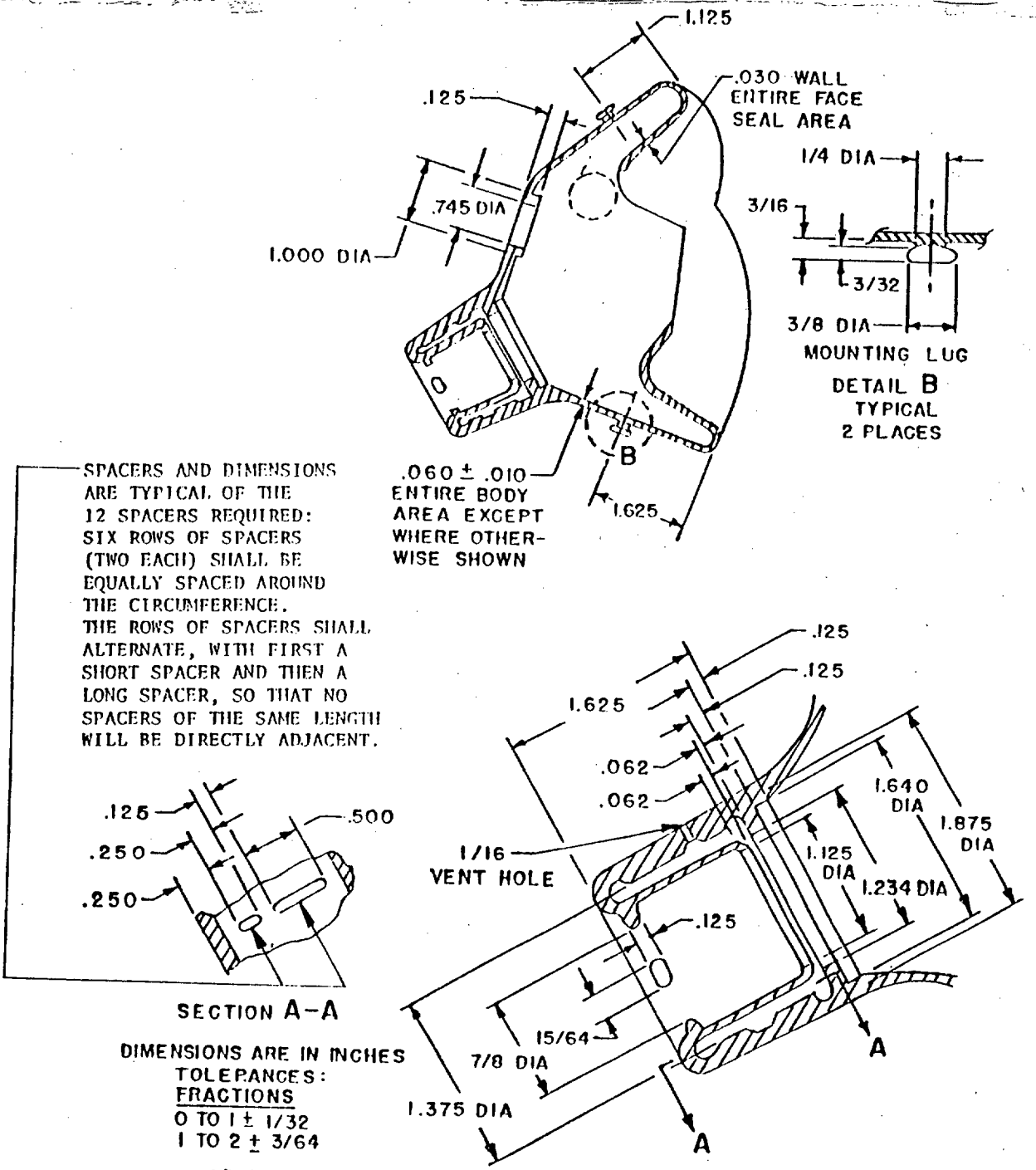


FIGURE 1. Facepiece

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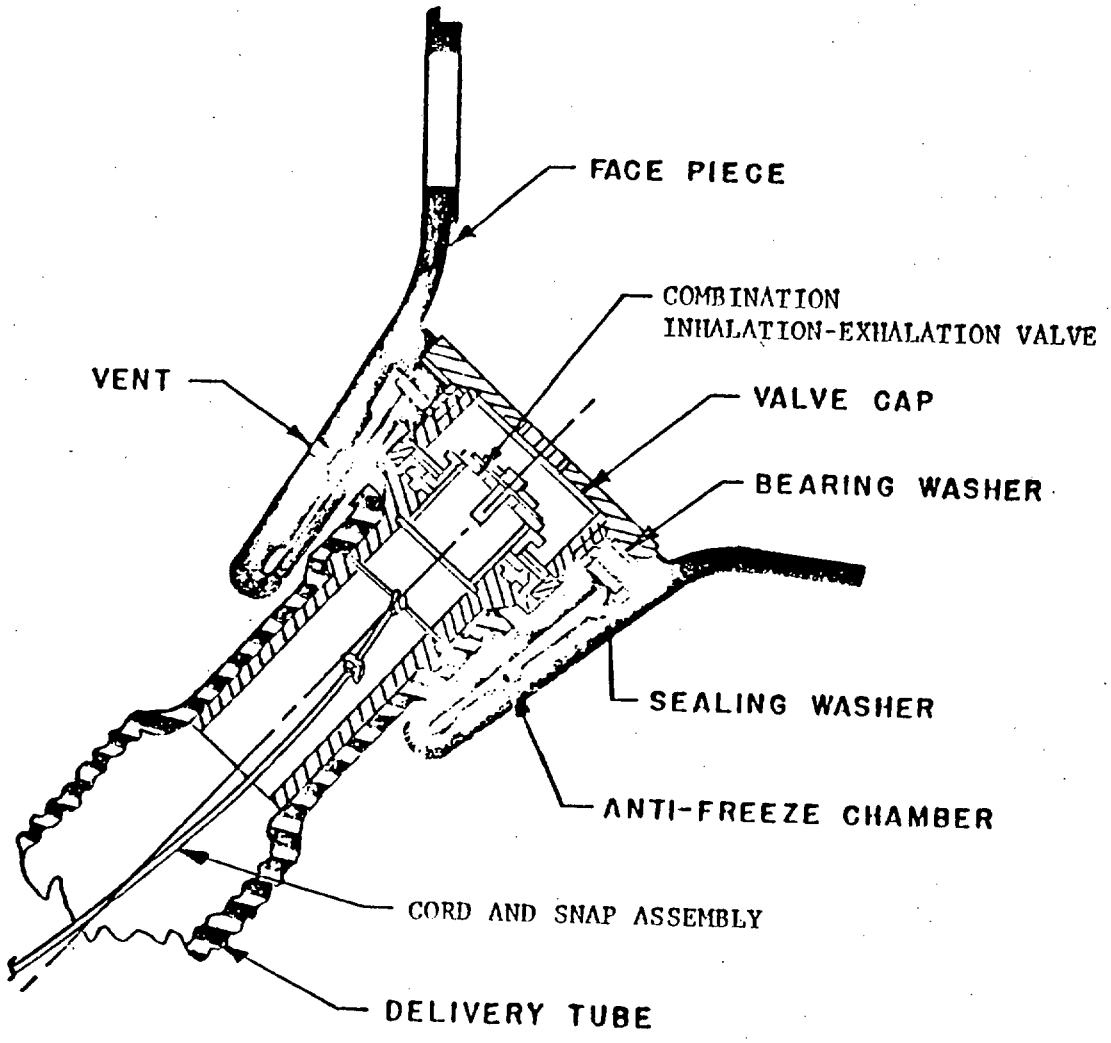
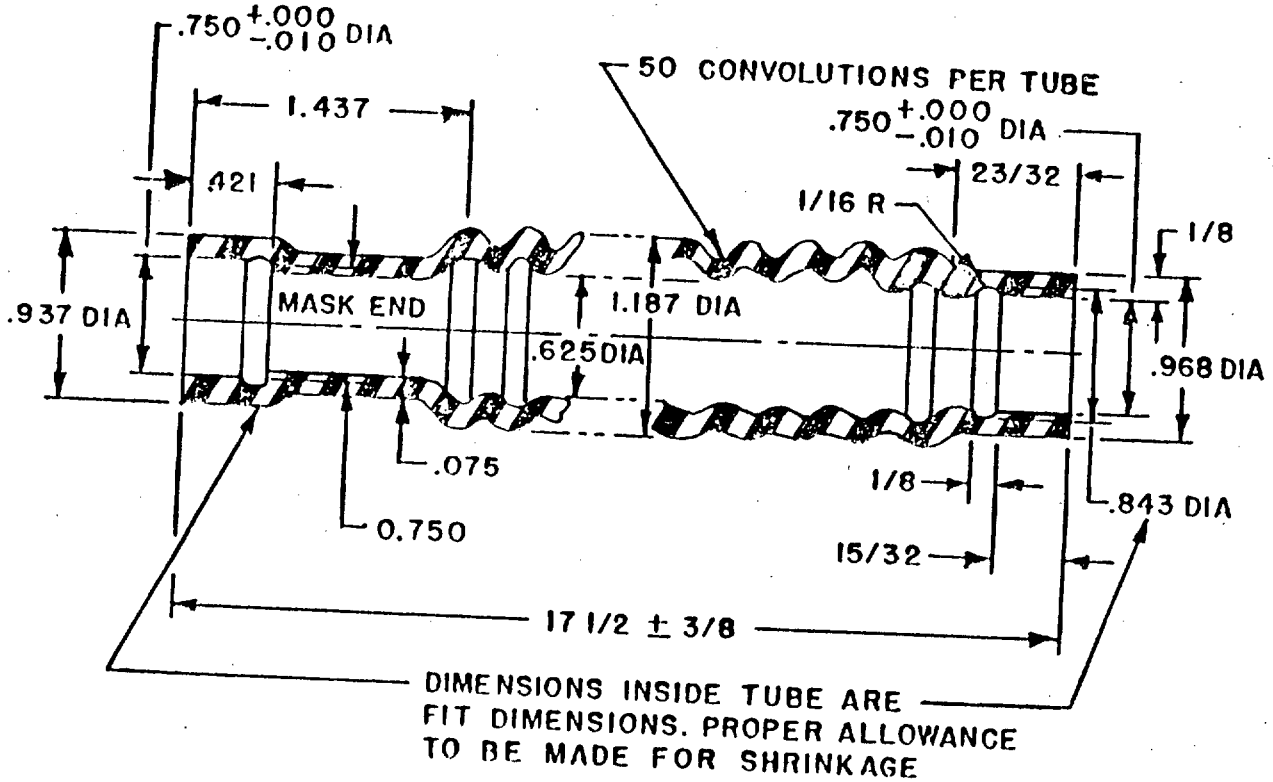


FIGURE 2. Valve installation

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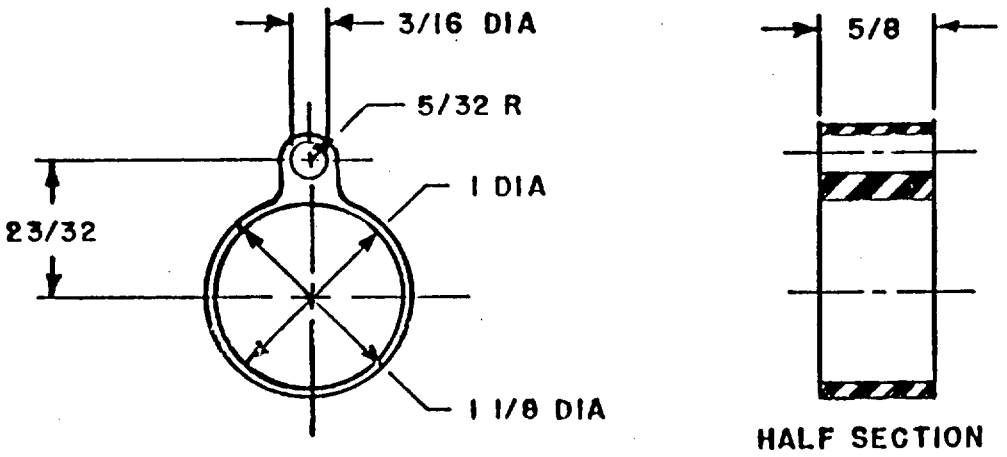
DIMENSIONS IN INCHES

TOLERANCES:

FRACTIONS	DECIMALS
± 1/32	± .010

FIGURE 3. Delivery tube

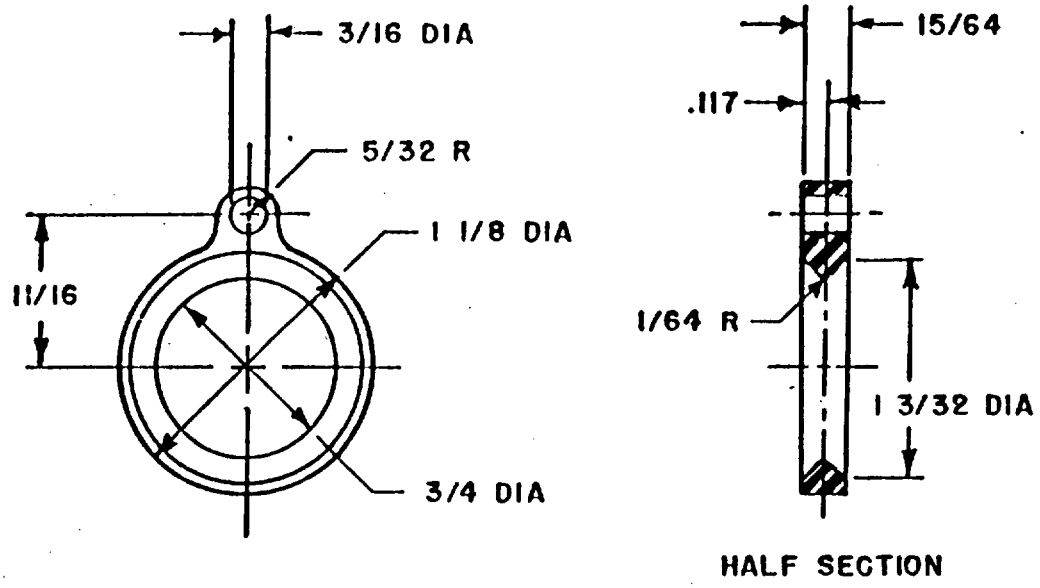
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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED, TOLERANCE FOR ALL DIMENSIONS ±1/64

FIGURE 4. Lower cable guide

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED, TOLERANCE FOR ALL DIMENSIONS ±1/64

FIGURE 5. Upper cable guide

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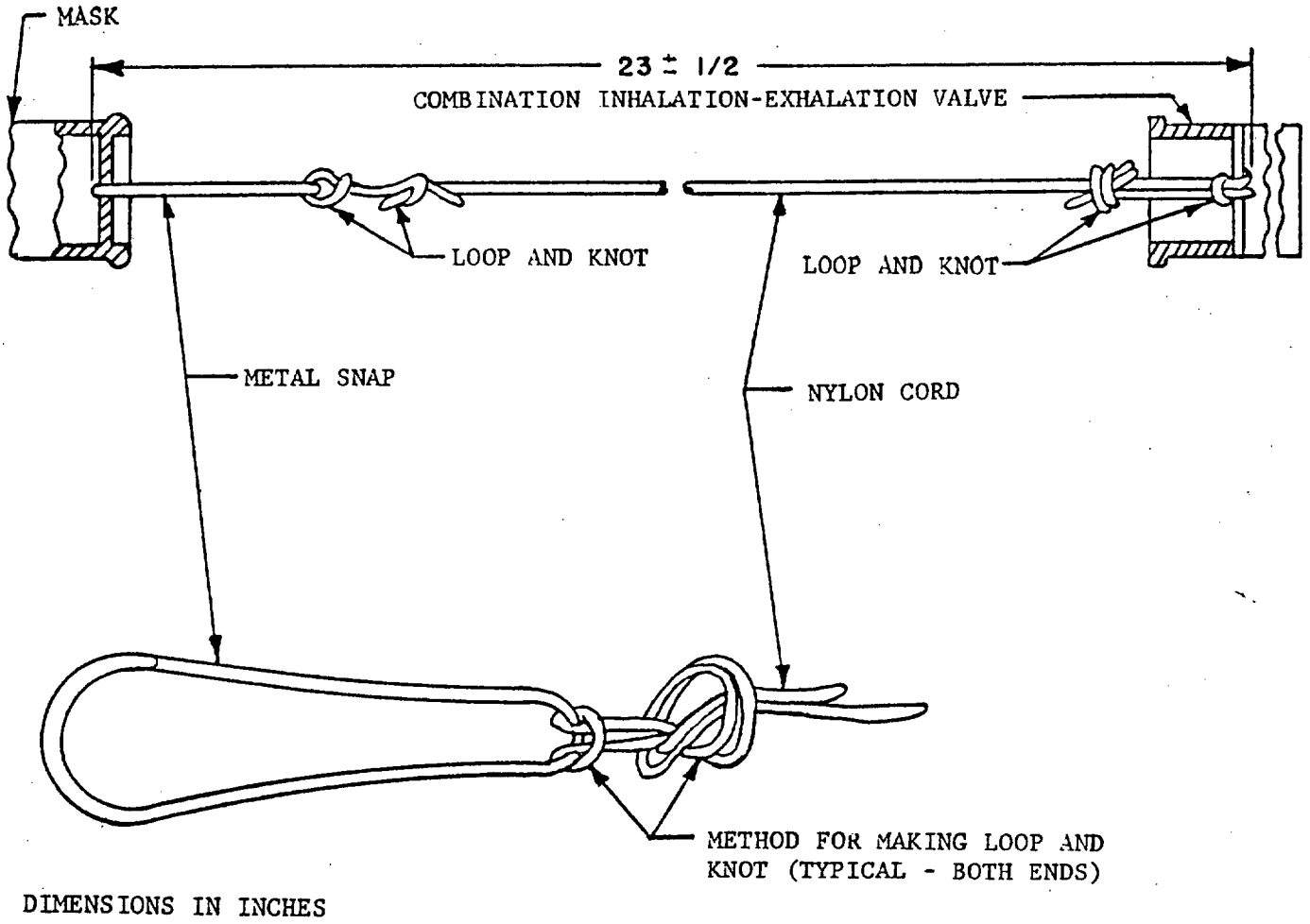


FIGURE 6. Cord and snap assembly

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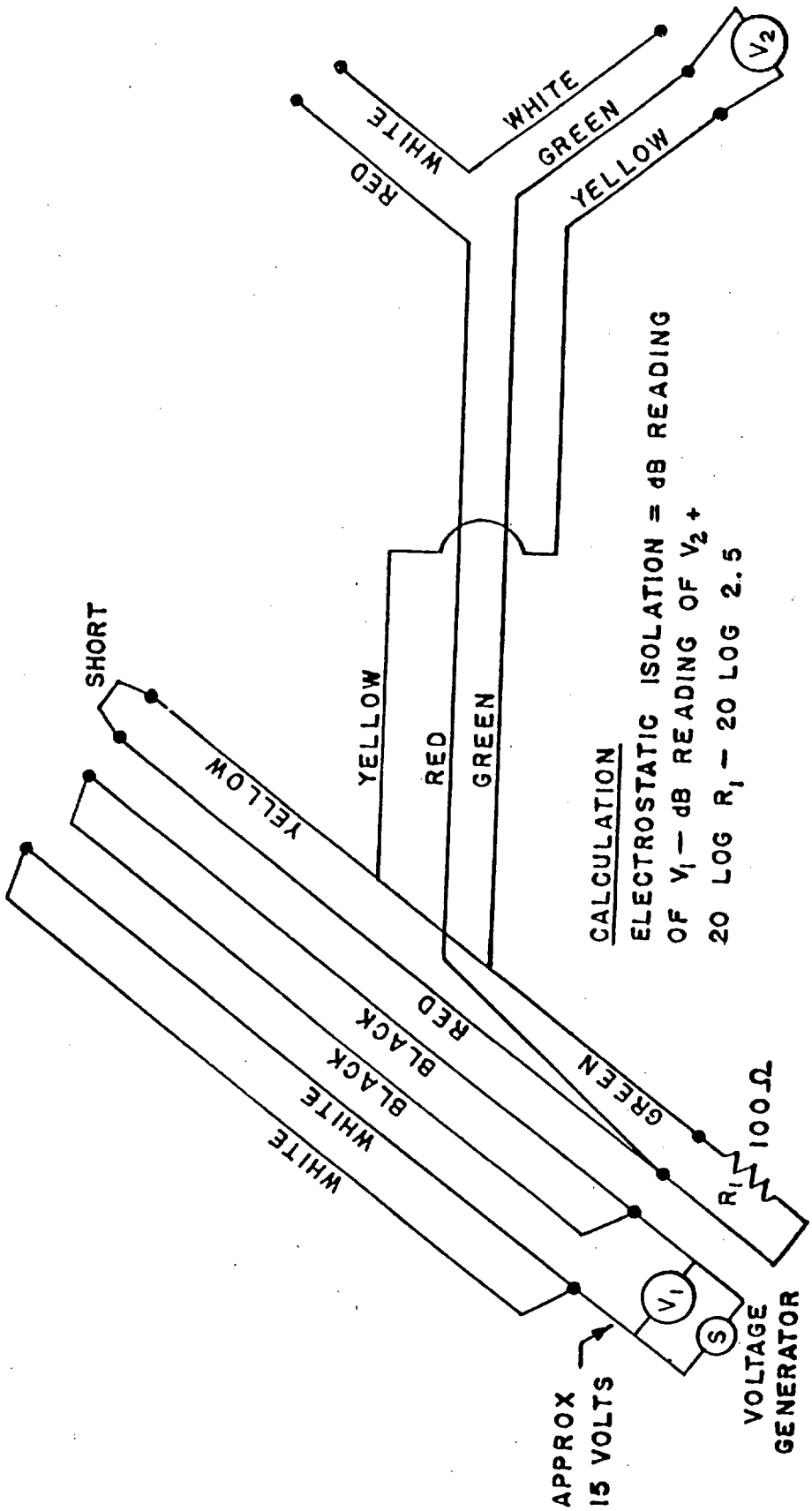


FIGURE 7. Electrostatic isolation test.

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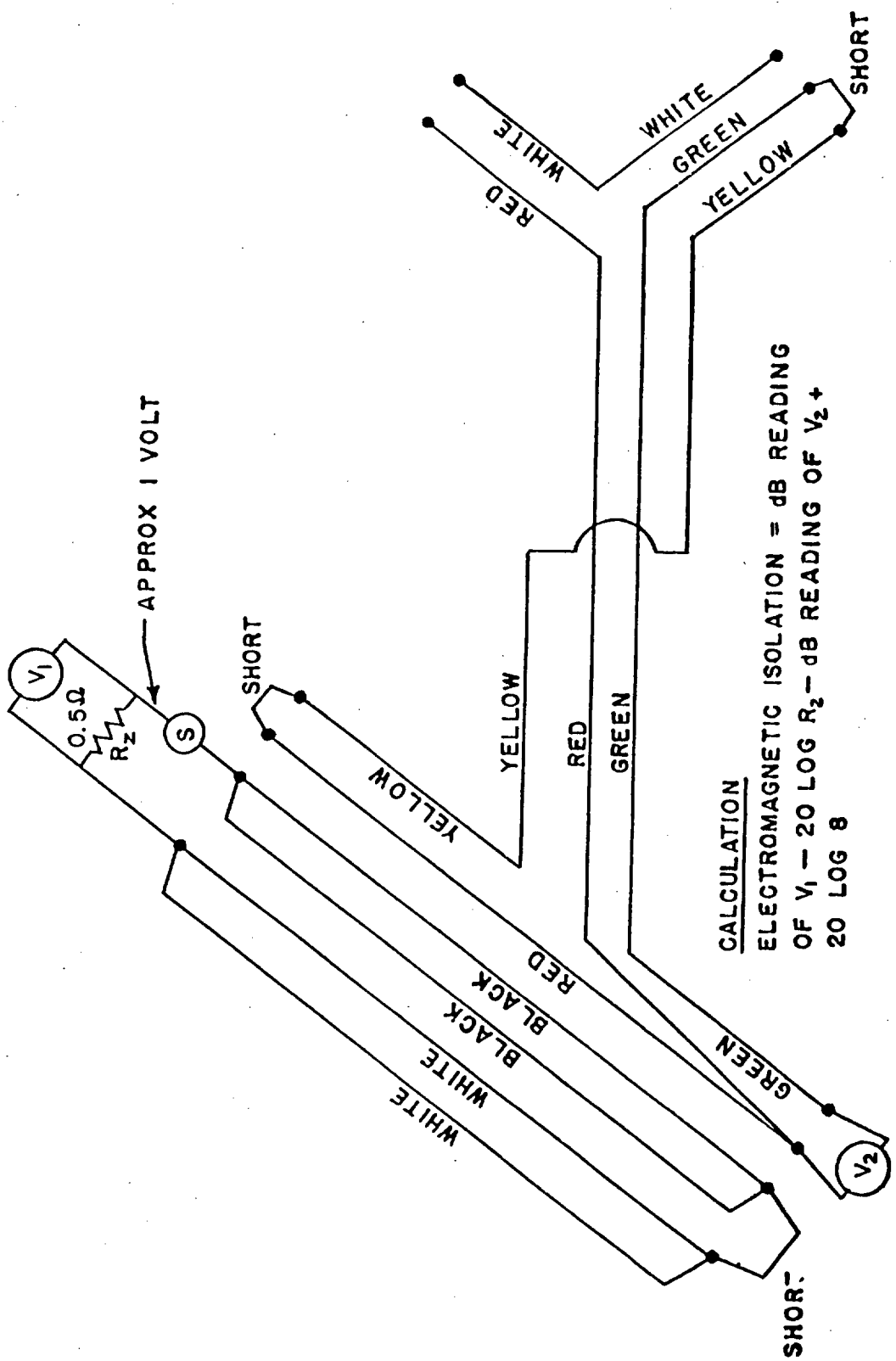


FIGURE 8. Electromagnetic isolation test.

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Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

Agent activity:
Air Force - 99

(Project 1660-F644)

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